

Description

The Residual Variation Intolerance Score (RVIS) “assesses the degree to which genes have either more or less common functional variation than expected for the genome as a whole given the amount of presumably neutral variation they carry”. This is based on the observation that for a gene with very few functional mutations in the general population, observing a functional variant in a patient would be very striking, but for a gene that often carry functional mutations at high frequencies the same observation would not be less striking.

Definition:

In gene i , let

Y_i : the number of common (Minor Allele Frequency (MAF) $>\rho$) functional variants

X_i : the total number of protein-coding variants (functional or non-functional, any MAF)

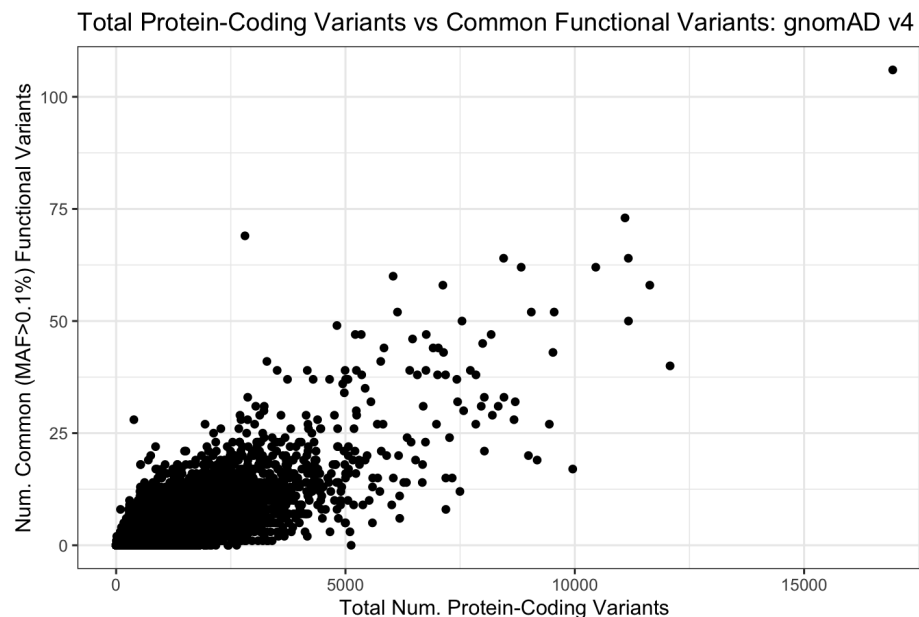
To compute RVIS, take a simple linear regression of Y on X :

$$Y \sim X$$

then take the studentized residual as the RVIS.

Steps

You will be getting the RVIS for a subset of genes on chromosome 19, 20, and 21, and using the scores to interpret the likelihood of the genes being disease-causing.



1. Download the dataset from https://github.com/gerhodes/bios710_FA26. The filename is `gnomAD_gene_counts_subset_RVIS_upd.tsv` and includes 3 columns:
 - (a) `gene`: HGNC symbol
 - (b) `num_common_functional`: count of functional variants with a minor allele frequency exceeding 0.1%

- (c) `total_coding_variants`: count of all protein-coding variants, functional or non-functional
2. Fit a regression model to the dataset and extract studentized residuals to get the RVIS value for each gene.
 3. Choose 2 of the following genes and find their RVIS value in your output:

SCN1A, *SETD1A*, *KMT2D*, *ARID1B*, *POGZ*, *OR10G4*, *TAS1R2*, *OR1A1*

- (a) Interpret the RVIS residuals for each of the genes. Would you consider them intolerant?
- (b) Use the gnomAD browser <https://gnomad.broadinstitute.org/> to look up the genes, and locate the constraint table in the upper right. Does the LOEUF agree with your RVIS value?
- (c) Using one of the following resources, look up the genes you chose. Do the function and/or disease associations for the gene make sense given what you've found?
 - <https://omim.org/>
 - <https://www.genecards.org/>
 - <https://www.ncbi.nlm.nih.gov/>